KONAMARTHI NITISH VEERA VENKATESH

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Education

Anil Neerukonda Institute of Technology and Sciences Sep 2022 – Jun 2026
B.Tech in Computer Science & Engineering *Visakhapatnam, India*

CGPA: 8.92/10.0 | **Coursework:** Data Structures & Algorithms, OOP, DBMS, Software Engineering, Cloud Computing

Certifications

- **AWS Certified AI Practitioner (AIF-C01)** – Generative AI, RAG, embeddings, vector databases [Badge]
- **AWS Academy Graduate – Cloud Foundations** [Badge]
- **OCI 2025 Certified Foundations Associate** [Badge]
- **OCI 2025 Certified AI Foundations Associate** [Badge]
- **Postman API Fundamentals Student Expert** [Badge]

Technical Skills

AI/ML Concepts: Generative AI, LLMs, RAG, Embeddings, Vector Databases, Prompt Engineering
Cloud: AWS (EC2, S3, Lambda, VPC, BedRock), Oracle Cloud Infrastructure (OCI)
Backend & APIs: Node.js, Fastify, Express.js, FastAPI, REST APIs
Languages: Java, Python, JavaScript, TypeScript
Databases: MySQL, PostgreSQL
Tools: Linux, Git, GitHub, Postman, VS Code

Projects

TokenSentry – AI Governance & Model Routing Layer for Coding Agents Expected Completion: Q4 2026
Node.js, Fastify, TypeScript *In Progress*

- Building a plugin-based proxy layer that sits between coding agents and upstream LLM APIs, intercepting requests to enforce usage policies and token budgets without changing the agent's workflow.
- Designing a cost-aware model routing engine that selects a model per request using provider, capability, and pricing metadata.
- Architecting the layer to be provider-agnostic, with governance logic kept separate from provider connectivity and API-key handling, so it works across different LLM providers.

English Speaking Tutor – Local-First Voice Tutor with Semantic Memory Currently Offline; AWS Deployment Planned
FastAPI, llama.cpp, Whisper, Kokoro, SQLite, sqlite-vec *In Progress*

- Built a fully local voice tutor on a 4 GB RTX 3050 laptop using Whisper-small ASR, Qwen3-4B via llama.cpp, and sentence-streamed Kokoro TTS over a FastAPI/WebSocket backend; no audio or transcripts leave the machine.
- Implemented hybrid memory retrieval combining SQLite FTS5 with all-MiniLM-L6-v2 embeddings (sqlite-vec) via reciprocal rank fusion (31.5 ms mean query latency); kept the cross-encoder reranker disabled until evaluation justifies its cost.
- Kept only the LLM on the GPU and ran ASR, TTS, and embeddings on CPU to avoid VRAM contention; validated with 17 automated tests and 9 frontend voice checks, with AWS deployment planned after reliability testing.

Chest X-Ray Diagnostic Assistant *FastAPI, DenseNet-121, Gemini API, Geoapify*

- Developed a DenseNet-121 deep-learning model to detect and classify 14 chest X-ray pathologies, achieving 87.8% accuracy.
- Built a FastAPI inference backend and integrated an LLM API to turn model predictions into structured, explainable reports.
- Implemented image validation and preprocessing ahead of inference to reject invalid diagnostic inputs.
- Deployed on Vercel and Hugging Face; integrated the Geoapify API to recommend nearby healthcare providers.

Achievements

- **Smart India Hackathon 2024:** Selected at national level for problem SIH1637 (Agriculture & Rural Development); collaborated with a cross-functional team to build an agri-tech prototype.
- **ANITS x PivotalSoft 24-Hour Hackathon 2024:** Built a full-stack prototype within 24 hours, demonstrating rapid prototyping and problem-solving.